Task#1: How are you gonna prompt? Share a structure for your prompts with your lead.

Prompt:

“Act as a Trainee Software Engineer. We are building a training DRF-based backend project "**DocSphere**". Write a Django class-based view to handle bulk user uploads via Excel files (.xlsx). Assume DRF’s abstract user already exists. Our target audience is fellow trainee software developers. Provide the output in a clean markdown code block following the PEP-8 compliance, good coding principles. Include error handling, proper validations and meaningful variable names along with API responses.”

Task#2: **Scope scenarios where we can employ each of the different advanced prompt engineering techniques.**

Advanced Prompting Techniques:

A. Step-by-Step Modular Decomposition:

1. **Chain-of-Thought (CoT):**

   Explain how neural networks work by breaking explanations into clear logical stages. Start from layers of neural networks including input, hidden and output layers. Also discuss forward and backward propagation. Discuss cost function with appropriate examples. Let's think about everything step-by-step in a clear, concise and easy-to-understand way.

   Zero-Shot Chain-of-Thought (CoT):

   Trainee Software Engineer receives a 32 hour deadline to accomplish a task. If they work 8 hours per day, how many days will it take to finish the task? Let’s solve this step-by-step.

   Few-Shot Chain-of-Thought (CoT):

   The password of the laptop changes automatically as the sum of the date, month and year.

   Example#1:

   Date: 26 January, 2026 → Password: 26 + 1 + 2026 = 2053.

Example#2:

Date: 22 February 2026 → Password: $22 + 2 + 2026 = 2050$.

What will be the password for 17 June, 2026. Clearly show reasoning in a step-by-step manner.

Automatic Chain-of-Thought (CoT):

I'm providing you a list of 10 different organization names from our DocSphere project.

Stage#1 (Clustering): Group permissions for these organizations based on their requirements.

Stage#2 (Demonstration): For each organization cluster, select representatives (like organization admins), and list their organization resources such as related projects and documents for each cluster. Let's think step-by-step and break down your data flow and logical reasoning for each step.

Stage#3 (Separation): Using the organization data demonstration from Stage#2, assign the appropriate roles permissions to organization users.

[excel file containing organizations list attached]

2. **Tree-of-Thought (ToT):**

   We are managing 10 different organizations for our DocSphere project.

   Goal: Assign appropriate role permissions to organization users.

   Instructions:

   1. At each stage (clustering → demonstration → separation), generate multiple reasoning paths and analyze how role-based permissions are organized.
   2. Evaluate each path for clarity, conciseness and correctness.
   3. Consider evaluating paths and continue branching until the most optimal and promising permission structure is determined.

   [excel file containing organizations list attached]

3. Graph-of-Thoughts (GoT):

We are organizing an annual dinner for Cogent Labs.

Goal: Find the most economical and suitable venue for the event.

Instructions:

1. Generate multiple reasoning nodes for each considerable venue option (indoor, outdoor, rooftops).
2. Connect the reasoning nodes where dependencies exist (e.g: weather conditions, distance from office, arrangement cost).
3. Evaluate each node for cost, feasibility and convenience.
4. Merge nodes where paths lead to similar conclusions.
5. Evaluate suboptimal nodes and continue exploring until the most optimal venue is identified.
6. Provide a graphical representation showing nodes, edges and final recommended venue.

B. Comprehensive Reasoning and Verification

1. Automatic Prompt Engineer (APE):

I want to write an official documentation focused on Generative AI in education. Generate 10 different versions by using knowledge from research articles and other authentic resources. Score them on the basis of clarity, accuracy, conciseness, completeness and alignment with industry standards. Output the best version by clearly highlighting the heading, executive summary and citing the references according to MLA-format.

2. Chain-of-Verification (CoVe):

Statement: The 2026 Cricket World Cup was held in Pakistan.

Verification Questions:

1. Have any Cricket World Cups been held in Pakistan before?
2. Is the 2026 Cricket World cup held yet?

Verification Answer: No. No Cricket World Cup has been held in Pakistan, and the 2026 World Cup has not taken place yet.

3. Self-Consistence:

A trainee Software Engineer is assigned a 32-hour task. They work 8 hours per day.

Instructions:

1. Solve this calculation 5 times.
2. Show step-by-step calculation for each attempt.
3. Identify the answer that appears most frequently.
4. Give the final answer in days.

4. **ReAct (Reason + Act):**
   A trainee Software Engineer is assigned a 32-hour task. They work 8 hours per day.
   Instructions:
   1. **Reasoning:** Think step-by-step about how to calculate the number of days based on total hours and per-day working hours.
   2. **Acting:** Perform calculations at each reasoning step.
   3. Verify calculations at each step.
   4. Report total number of days required to complete assigned task.

C. Usage of External Tools/Knowledge or Aggregation

1. **Active Prompting:**
   Write a DRF view for nested AbstractUser for our training backend-project DocSphere.
   Instructions:
   1. If there exists any ambiguity in assigning permissions, mark it as *Unclear Instructions*.
   2. Include proper validation checks and comments in code wherever you find any uncertainty so that I can provide a manual example.

2. **Automatic Multi-step Reasoning and Tool-use (ART):**
   We are traveling from Lahore → Islamabad → Murree with various flight and train options available. Find the fastest route by considering the travel time required and commute cost.
   Instructions:
   1. Break the problem into segments: Lahore → Islamabad, Islamabad → Murree
   2. Use all the available tools: distance calculator to calculate the distance, expense calculator to compute the commute cost, analyze flights and trains timetables.
   3. Compare different alternative options automatically.
   4. Provide a segment-by-segment plan showing chosen transport, travel time and cost.
   5. Report the final most optimal travel plan along with total travel time.